Investor Behavior

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Overview

- We want to understand investor behavior
  - the decisions investors make about their assets and liabilities
  - the portfolios they hold and how they trade

- This is an important topic:
  - There is a real danger that many investors are making poor financial decisions
    - Financial markets are complex and sometimes counter-intuitive
    - Many decision-makers are unsophisticated, and all the more so in recent years
  - If some people **are** making poor decisions, we need to find ways to help them do better

Overview

- We divide investors into individuals and institutions
  - focus more on individuals
  - they are more likely to be making poor decisions
- The field that studies the financial decisions of individuals is called “household finance”
  - It has grown rapidly in recent years

Financial decisions of individuals

- Assets (pre-retirement)
  - under-saving
  - non-participation
  - buying high, selling low
  - preference for active management
  - under-diversification
  - stock-picking performance
  - the selling decision
  - the buying decision
- Assets (post-retirement)
  - under-annuitization
- Liabilities

Under-saving

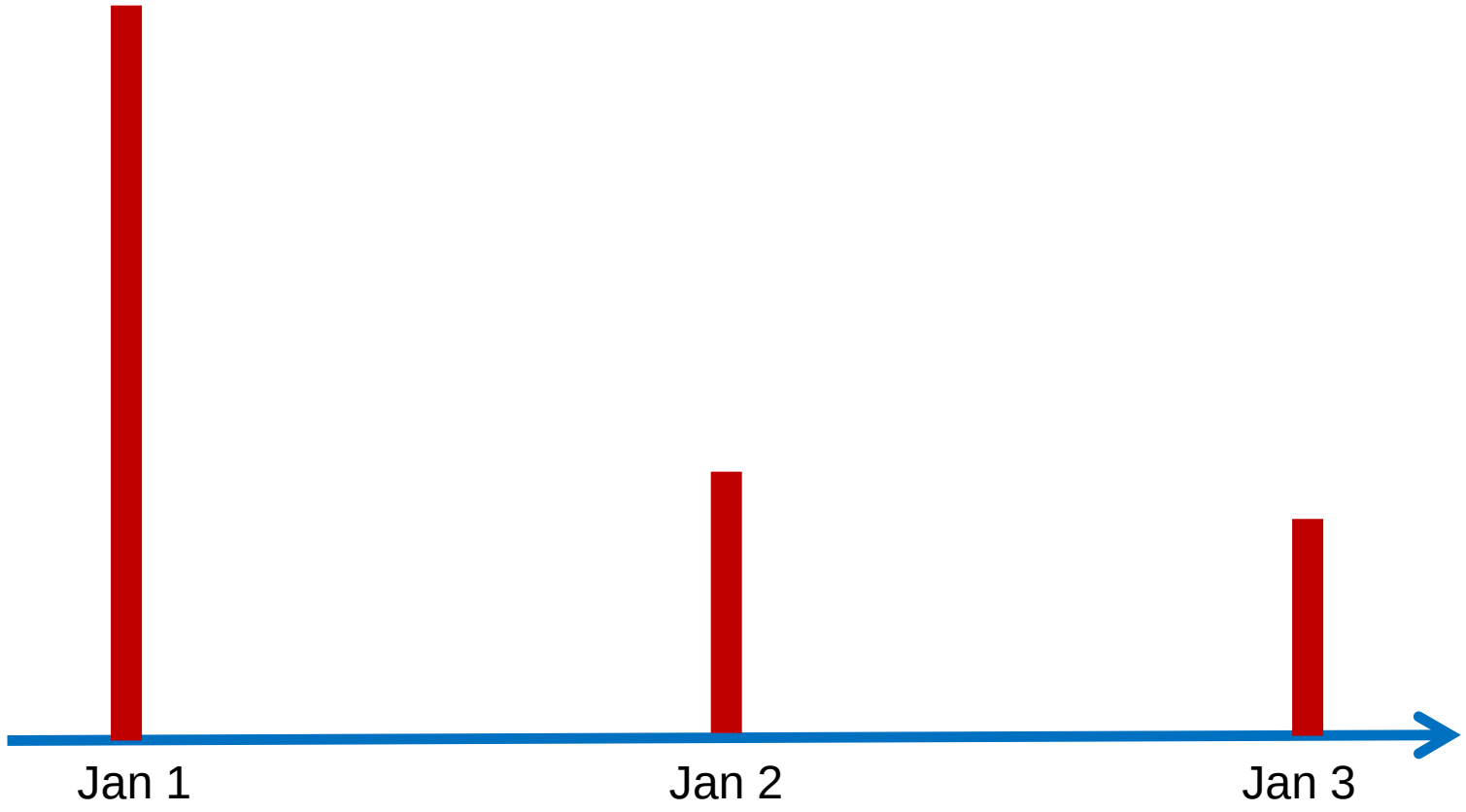
- Survey evidence suggests that many Americans are not saving as much as they should for retirement
- E.g., in a survey of hundreds of employees at a large U.S. company
  - two-thirds said that their saving rate was below the ideal
  - one-third said that it was about right
  - no one said they were saving too much
- Of those who said their saving rate was too low, 35% planned to raise it in the future
  - However, hardly any of them actually did

Behavioral approach: present bias

- Our heavy focus on the present leads us to consume too much today
  - and to save too little for the future
- Present bias also explains why people fail to follow through on their plans

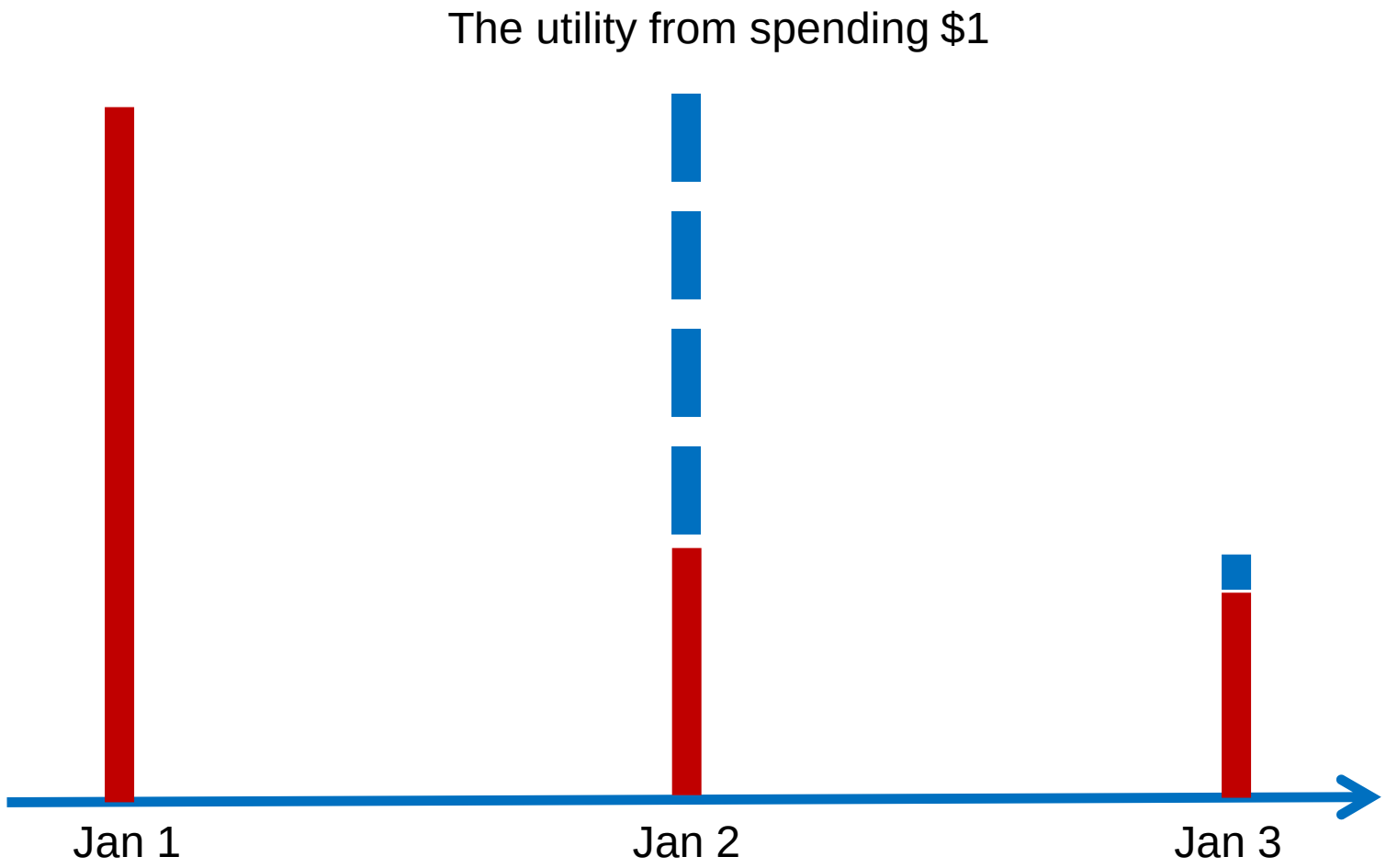
Present Bias

The utility from spending $1



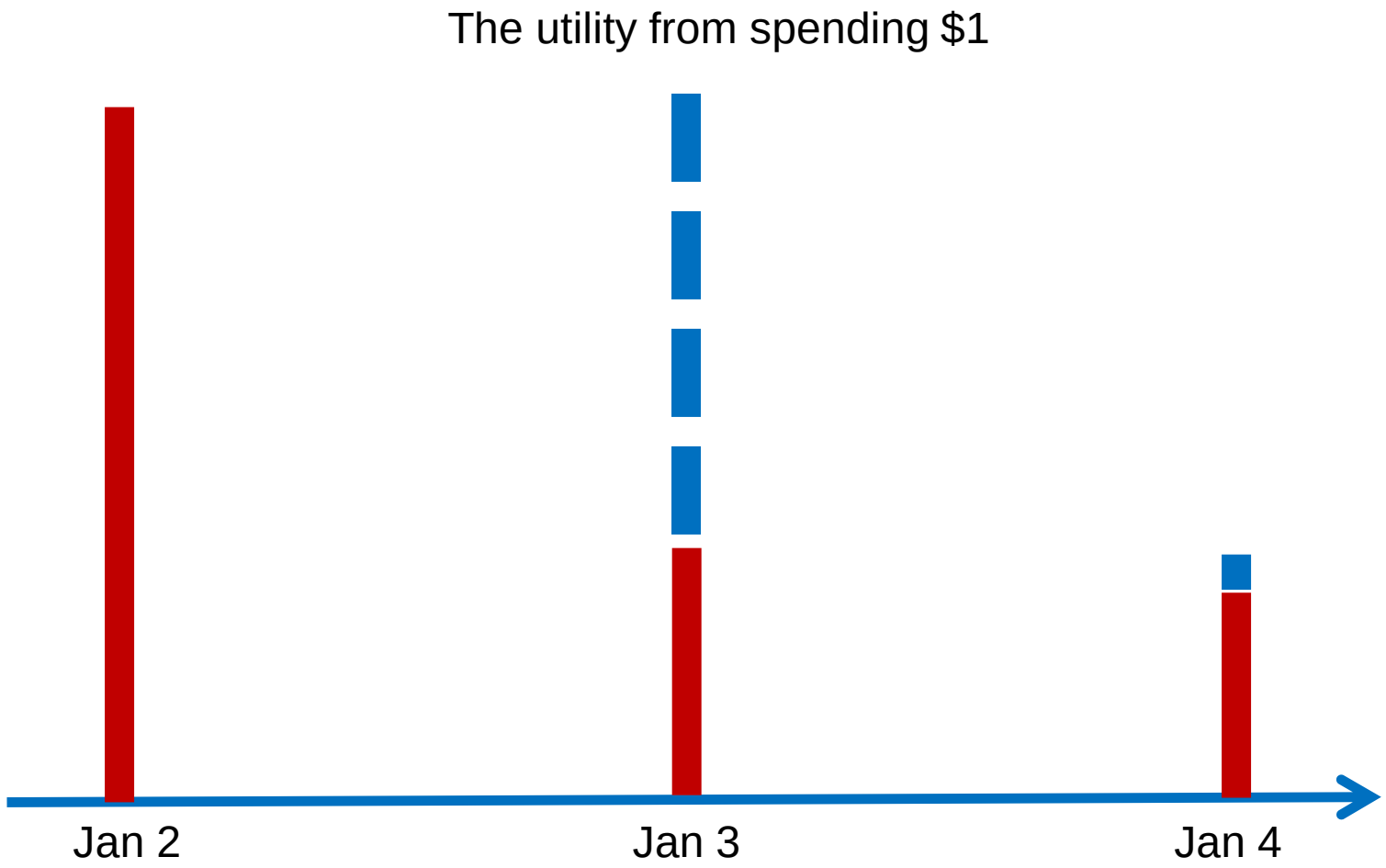
I will save (exercise, study, …) today or tomorrow?

Present Bias



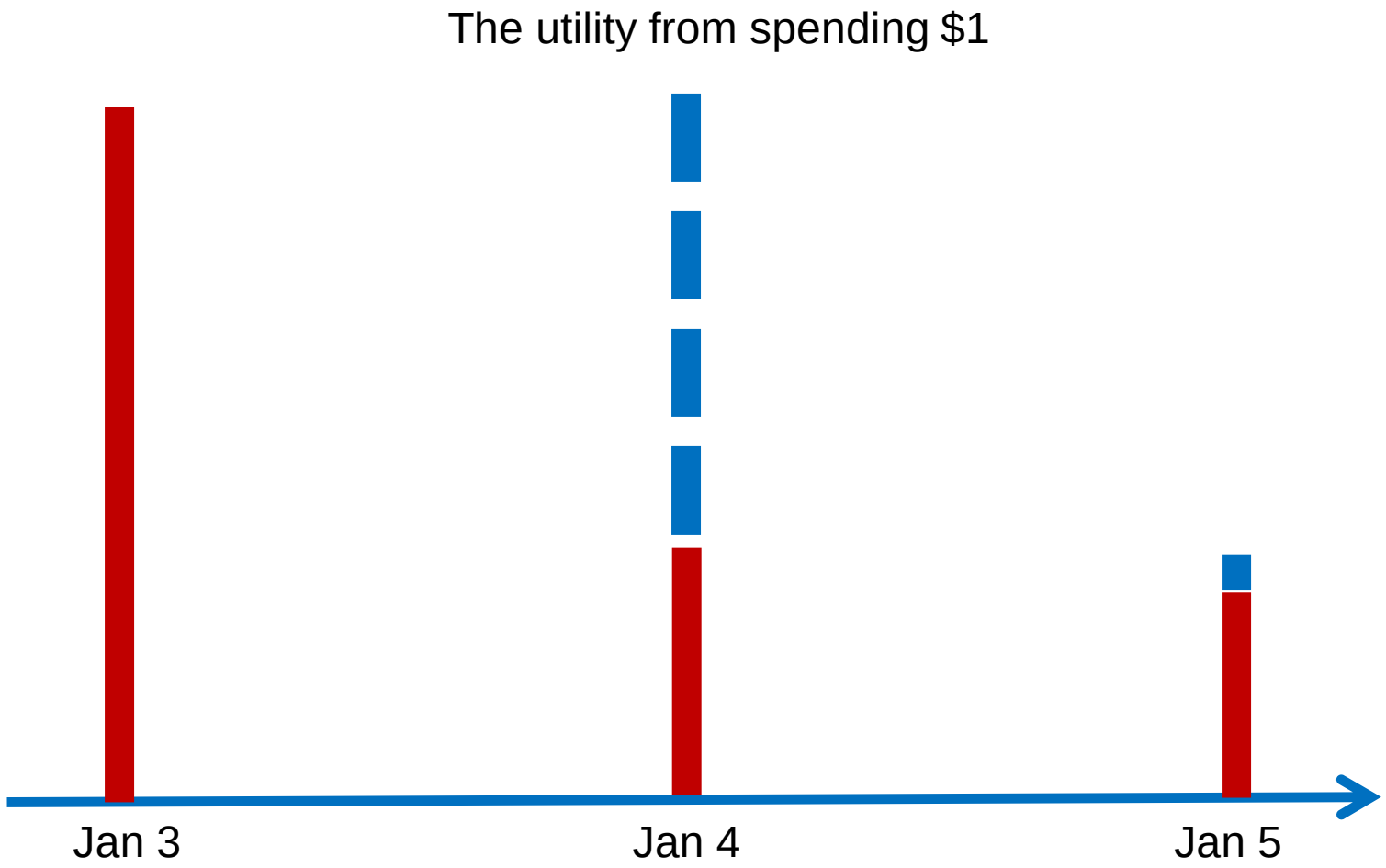
I will save (exercise, study, …) on **Jan 2**

Present Bias



I will save (exercise, study, …) on **Jan 3**

Present Bias

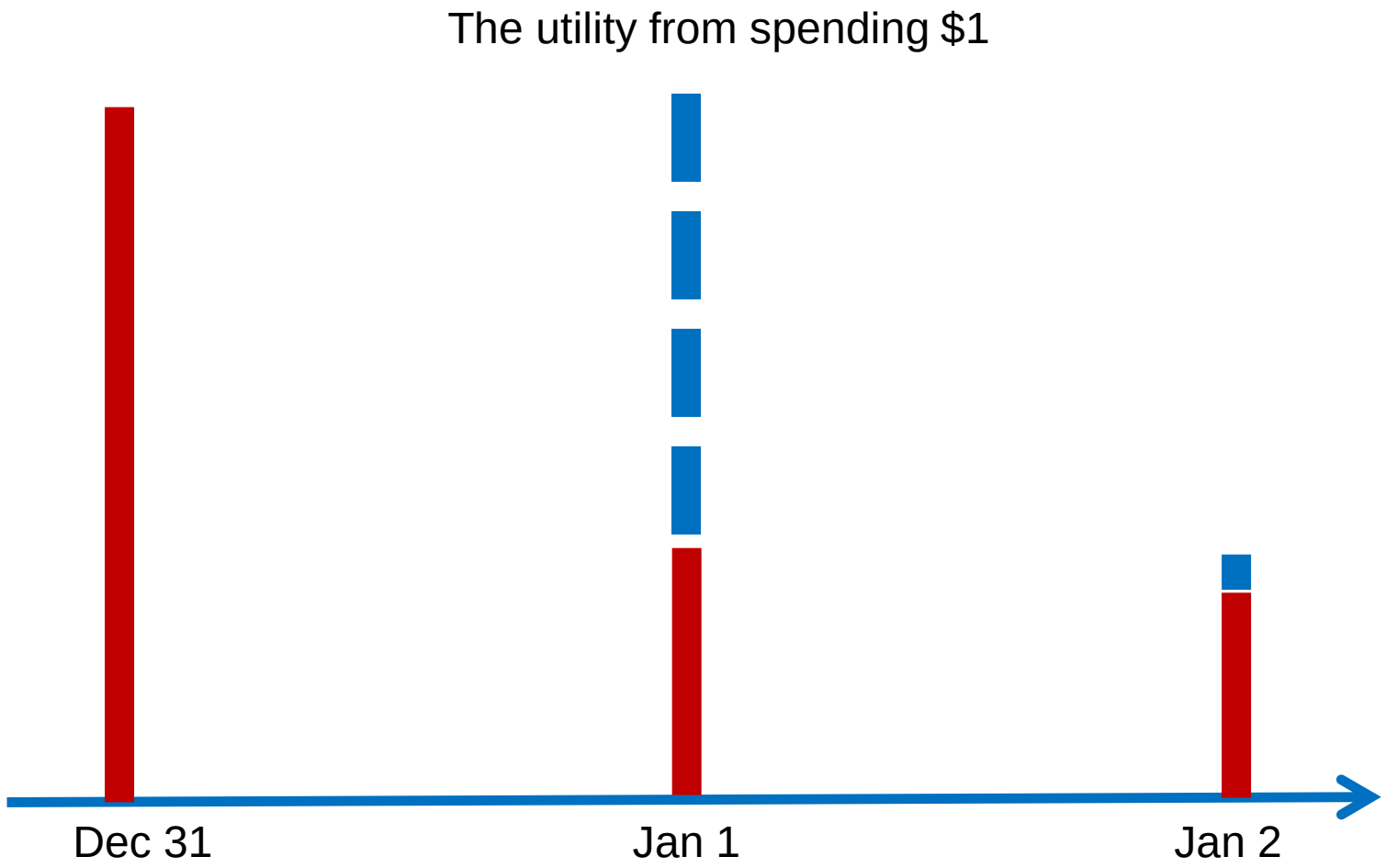


I will save (exercise, study, …) on **Jan 4**

Present Bias



Present Bias



I will save (exercise, study, …) on **Jan 1**

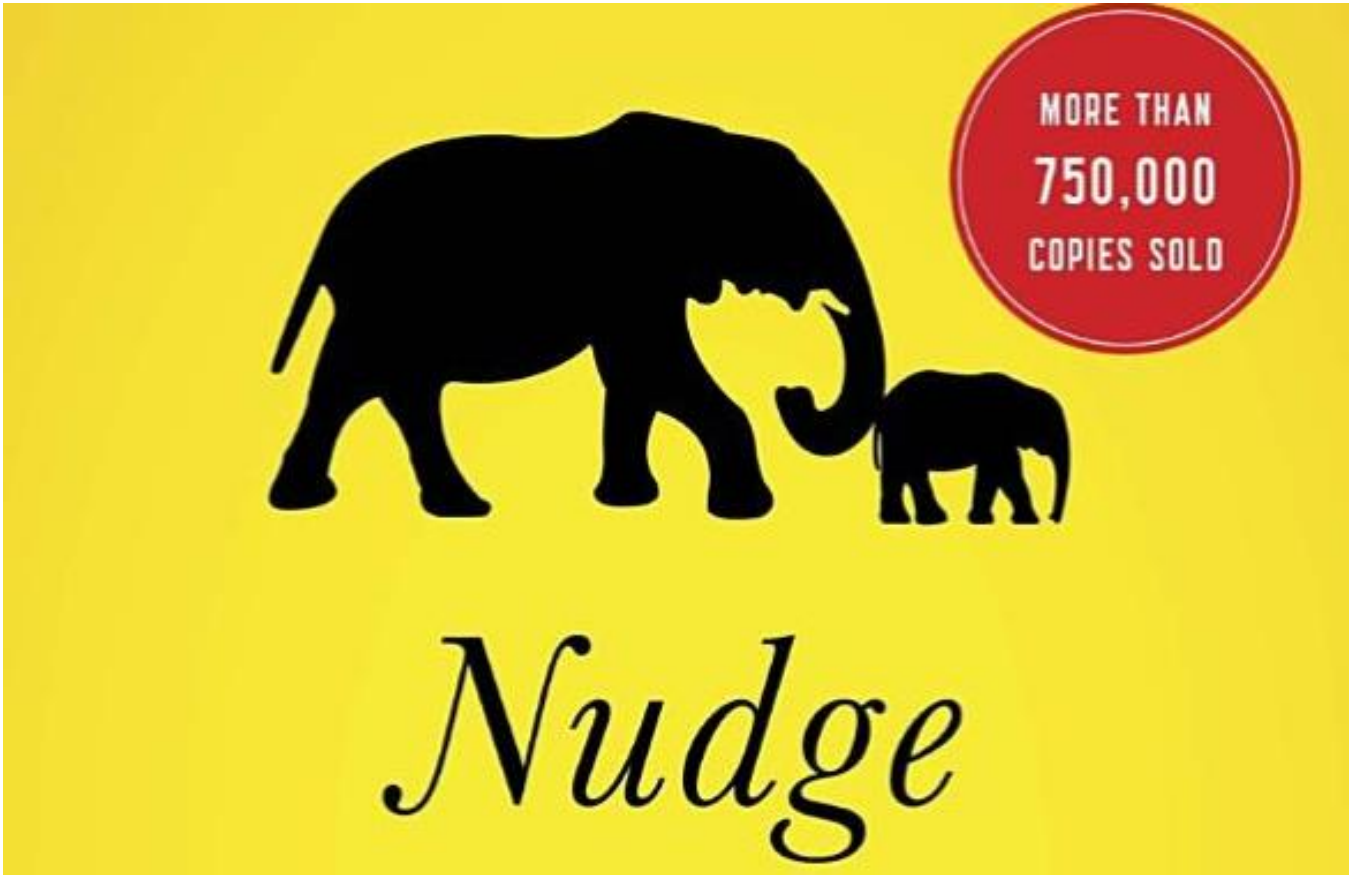
Implications of present bias

- Insufficient saving for pensions. We discount value of future pension payments and save insufficiently
- Consumption of demerit goods. We ignore future health costs for short-run utility of enjoying the drug.
- Delaying decisions. Given a choice between doing a project now, when we have time, we may put off until last moment, when we are tired and can't do as well.

Present Bias

- What am I doing?
- How can I deal with it? How can you help others?

Responses to present bias



Responses to present bias

- Nudges. Governments/pension companies may give us incentives to nudge us into taking out pension commitments.
- Highlight of costs. Government campaigns to highlight costs of consuming demerit goods or taxes to make them less palatable.

Non-participation

- A sensible investment portfolio should contain at least some exposure to the stock market
- For many decades, however, most U.S. households had no exposure at all to the stock market
  - e.g. in 1984, only 28% had any exposure
  - even among people with more than $100,000 in liquid assets, only 50% held any equity
- Even in more recent years, many households do not participate in the stock market

Non-participation

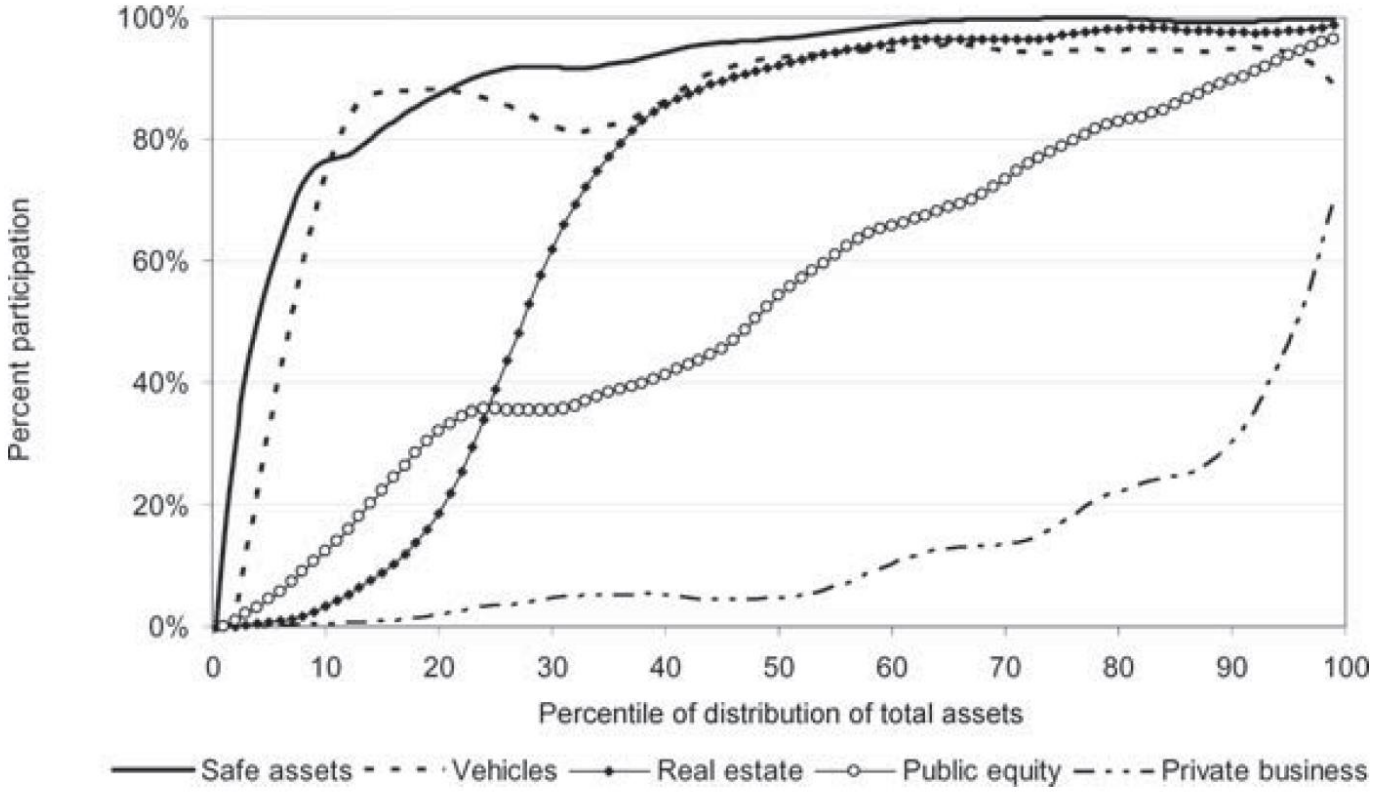


Figure 2. Participation rates by asset class. The cross-sectional distribution of asset class participation rates in the 2001 Survey of Consumer Finances.

Non-participation

- Rational approaches?
- Behavioral approach: prospect theory
  - Over the next year, the stock market may go up in value, but it may also go down
  - Investors who are loss averse find this prospect unappealing
  - Probability weighting makes the stock market even less appealing: (overweight small probability of crash)
    - Market return is negatively skewed

Non-participation

- Behavioral approach: **experience effects**

- When making a decision, we should take into account of **all** available data
- However, people appear to pay too much attention to the data that they have **personally experienced**
  - predicts that whether people participate in the stock market will depend on the stock returns they have experienced during their lifetime
- The data confirm this prediction
  - Malmendier and Nagel (2012)

Non-participation

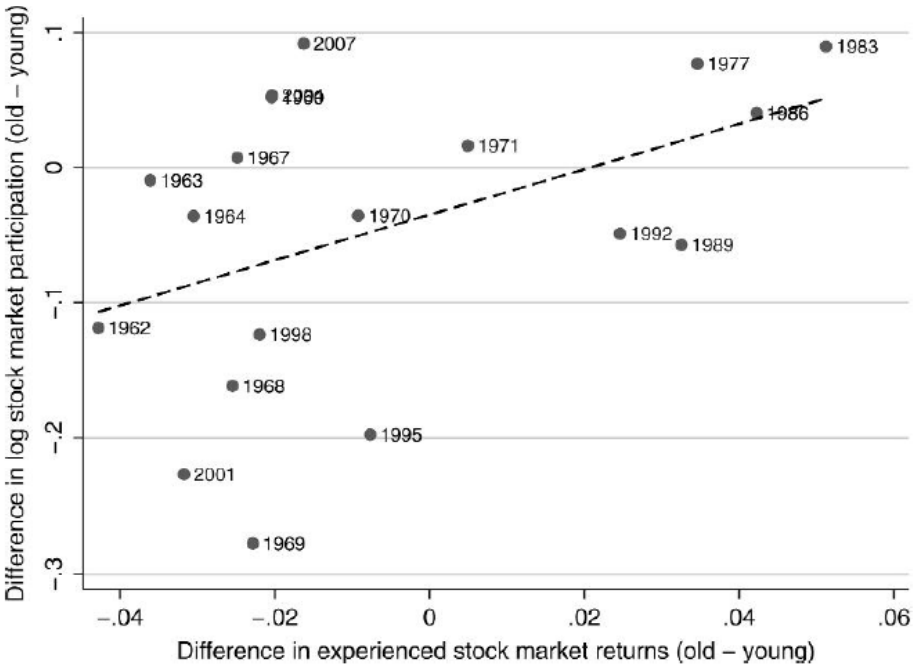


FIGURE I

Differences in Log Stock Market Participation Rates of Old and Young Individuals Plotted against Differences in Experienced Stock Market Returns

Stock market participation rates are the fraction of households who invest in stocks, including stock mutual funds and stocks held in retirement accounts. The vertical axis shows the difference in the log participation rates of old (household head age $>$ 60 years) minus young (household head age $\leq$ 40 years) households. The horizontal axis shows the average real stock market return (S&P 500 index) over the prior 50 years (as proxy for the return experienced by old households) minus the return over the prior 20 years (as proxy for the return experienced by young households). The years refer to the respective SCF survey waves. Observations are weighted with SCF sample weights. The 1960 and 2004 observations overlap in the plot.

Source: Malmendier and Nagel (2011)

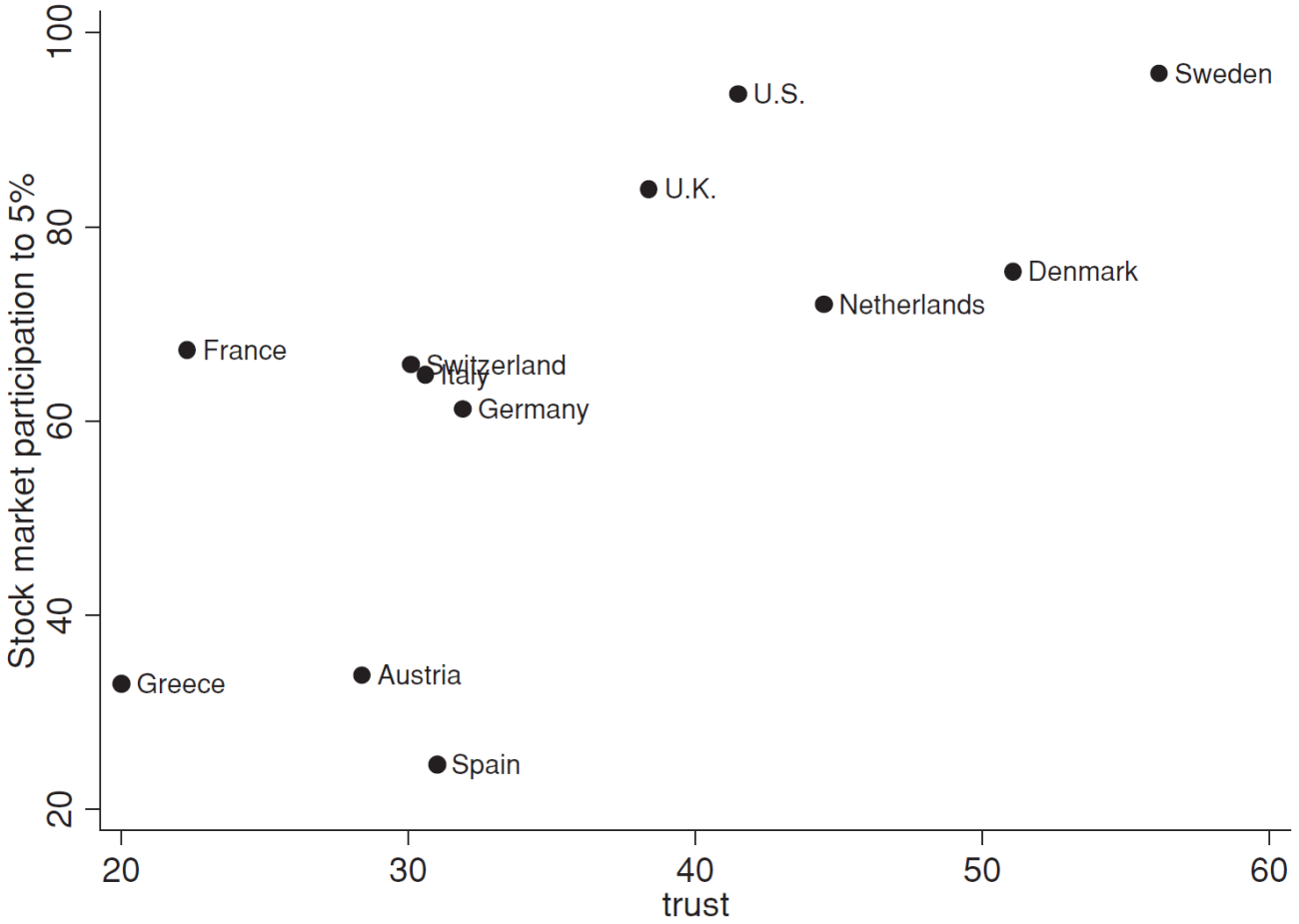
Non-participation

- Behavioral approach: **Trust**
- An investment in financial markets requires trust
- Trust can be measured through questions like:
  - "Generally speaking, would you say that most people can be trusted, or that you have to be careful in dealing with people?"
- In a study of 2,000 Dutch households, less trusting households were less likely to participate in the stock market
  - Guiso, Sapienza, Zingales (2006)

Non-participation

- Trust may also explain another puzzle: the difference across countries
- Trust explains the differences in participation rates across different countries
- More trusting countries have more participation

Non-participation



More on Trust

- Trust may also affect the composition of investors' portfolios
- Less trusting households tilt their portfolios more toward *dividend-paying* stocks (Kelly, 2014)

Correlation vs. causality

- Observation: when A happens, B is likely to happen
- Conclusion:
  - A caused B
  - B caused A
  - C caused both A and B.

- Trust and participation
  - Giannetti and Wang (2016) suggest that trust causes participation
  - Exogenous shock: The collapse of Arthur Andersen
  - Personal experience

Non-participation

- Behavioral approach: ambiguity aversion
- The stock market is more “ambiguous” than a bank deposit or T-Bill
- Investors who are ambiguity-averse therefore find the stock market unappealing

Buying high, selling low

- Many investors appear to have poor market timing
  - They increase their exposure to the stock market when it is highly valued, in advance of poor returns
  - and reduce their exposure when its valuation is low, in advance of high returns
- Key evidence: the **dollar**-weighted average return on mutual funds is lower than the **time**-weighted average return
  - by 1.5% per year
  - Investors chase past returns of mutual funds

Buying high, selling low

- Behavioral approach: **over-extrapolation**
- Some investors overweight past returns when forming beliefs about future returns

Preference for actively-managed funds

- Mutual funds come in two types: index funds and actively-managed funds
- A large body of research indicates that, after fees, actively-managed funds underperform index funds, on average
  - It is therefore puzzling that actively-managed funds are as popular as they are
- Rational approaches?
- Behavioral approaches?
  - Over confidence
  - Extrapolation

Under-diversification

- Traditional advice tells people to diversify their holdings of financial assets
- Many households appear to ignore this advice,
- Four types of under-diversification
  - home bias
  - local bias
  - concentrated stock holdings
  - large holdings of own-company stock

Home bias

- Individuals often invest heavily in **domestic** stocks, apparently ignoring the benefits of international diversification

- Rational approaches?
  - Information advantage
- Behavioral approach: Preference for the familiar/mere exposure effect
  - Mere exposure to something makes us like it more than justified based on informational considerations alone
  - We are exposed, every day, to our home country

Local bias

- Within their domestic equity holdings, individuals tilt toward locally-headquartered stocks
  - The average distance to a stock in the typical investor's portfolio is 917 miles
  - But the average distance to a stock, across all stocks, is 1225 miles
- Investors appear to be forgoing the diversification benefits of a more geographically-dispersed portfolio

Local bias (cont'd)

- Rational approaches?

- Behavioral approach: Preference for the familiar / mere exposure effect

  - We are exposed, every day, to our local region

Concentrated stock holdings

- Some households hold concentrated positions in relatively few stocks
- Rational approaches?
- Behavioral approach: **Overconfidence**
  - Investors may be overconfident about the validity of their analysis

Concentrated stock holdings (cont'd)

- Behavioral approach: **Prospect theory (probability weighting)**
  - People may want to have large positions in "lottery-type" stocks so as to give themselves a chance of becoming wealthy
  - There is empirical evidence to support this idea (Boyer, Mitton, Vorkink, 2010)

Large holdings of own-company stock

- People invest a surprisingly large fraction of their retirement savings in the stock of their own company (Benartzi, 2001)
  - 23% of their discretionary contributions
  - Worst under-diversification
- Rational approaches?
- Behavioral approaches?
  - Familiarity
  - Overconfidence
  - Loyalty
  - Inertia

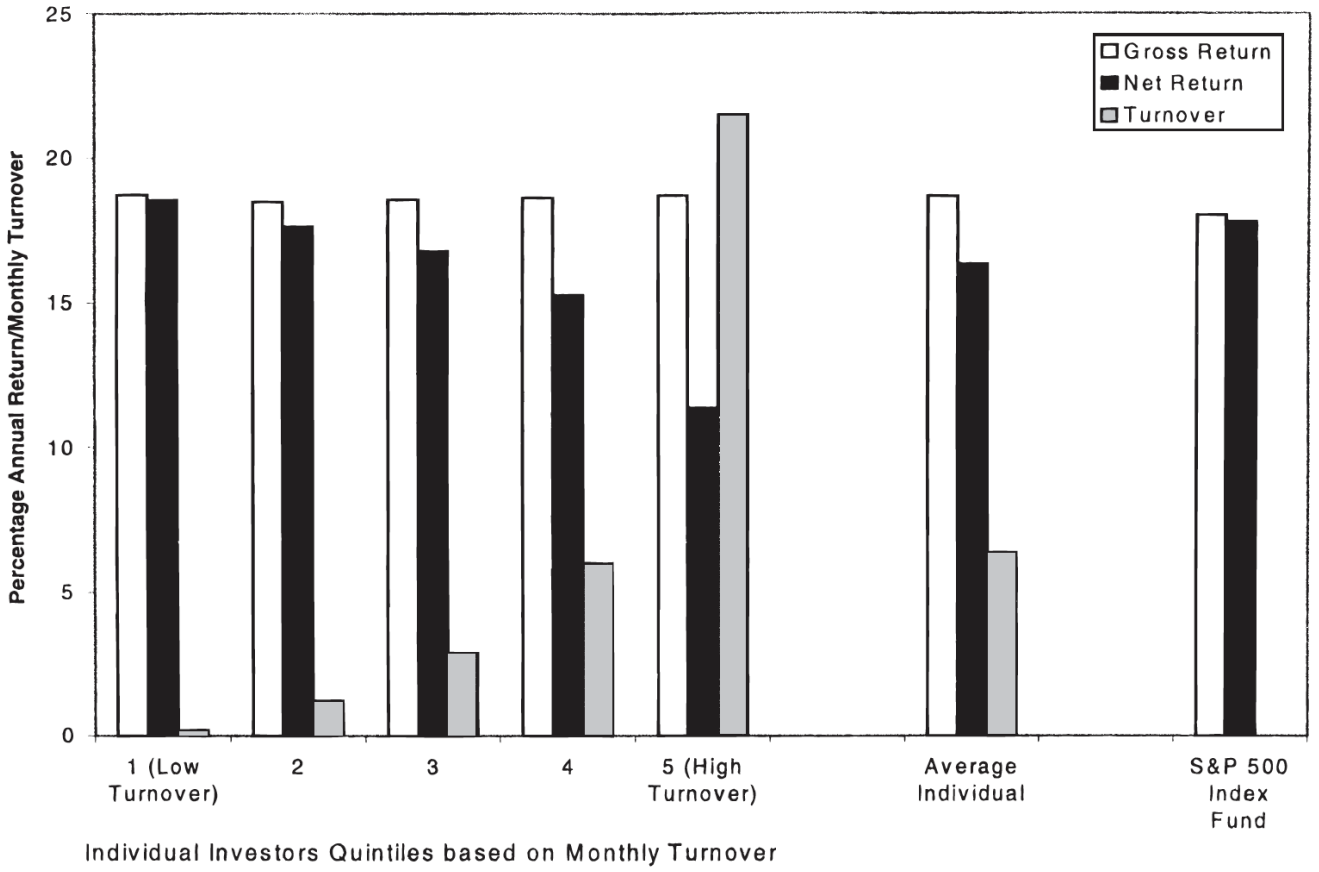
Stock-picking performance

- Most studies find that individuals do a poor job trading stocks

- They underperform a range of benchmarks
  - their beginning-of-the-year portfolio
  - the return of the overall market
  - a CAPM benchmark

Example: Barber and Odean (2000)

- Looked at the trading of 70,000 households through a large discount brokerage from 1991 to 1996



Stock-picking performance (cont'd)

- Rational approaches?
- Behavioral approach: **overconfidence**
  - Most investors believe that they are "above average" traders, and therefore trade stocks actively
  - Many are over-estimating their ability, however, and end up under-performing
- More Evidence?

Stock-picking performance (cont'd)

- More evidence:
  - This view predicts that more overconfident people will trade more
  - One study tests this using data from Finland
    - measures overconfidence as self-reported confidence minus an objective measure of confidence
    - finds that more overconfident people indeed trade more
  - The psychology literature suggests that men are more overconfident than women
    - this predicts that men will trade more, and perform worse
    - the evidence is consistent with this (Barber and Odean, 2002)

More on Overconfidence

- Overconfidence has also been applied to understanding the puzzlingly high level of trading volume in many financial markets

- In an economy where all investors are fully rational, there will be relatively little trading
  - Each investor will infer a negative signal from others' willingness to trade with them
- Trading volume will be much higher, however, when investors are overconfident
  - They disregard the signal in others' willingness to trade

Selling behavior: The disposition effect

- Individual investors have a greater propensity to sell a stock at a **gain** relative to purchase price, rather than at a loss (Odean, 1998)

- A similar effect has been documented in the real estate market
  - People appear reluctant to sell their homes below purchase price (Genesove and Mayer, 2001)

Selling behavior: The disposition effect

- (Potential) rational approach: **information**
  - Perhaps individuals sell past winners because they have good information that these stocks will subsequently do poorly
  - and hold on to their past losers because of information that these stocks will do well
- It turns out, however, that the past winners people sell subsequently do **better** than the past losers they hold on to
- Other rational approaches?

Selling behavior: The disposition effect

- Other evidence also suggests that the disposition effect is a mistake
  - Among individual investors, it is more pronounced for the “less sophisticated”
  - It is also more pronounced for individuals than for mutual fund managers

Selling behavior: The disposition effect

- Behavioral approach: **realization utility**
  - The idea that people feel pleasure when they sell an asset at a gain and pain when they sell an asset at a loss
- This simple idea was not taken seriously until recently
- Realization utility may stem from the way people think about their investing history
  - as a series of investing **episodes**, each one defined by the name of the asset, the purchase price, and the sale price

Selling behavior: The disposition effect

- Selling at a gain brings pleasure because it creates a positive investing episode
- Selling at a loss brings pain because it creates a negative investing episode
- Realization utility can explain the disposition effect, although an additional ingredient is needed
  - e.g. impatience (time discounting)
  - Barberis and Xiong (2012)

Selling behavior: The disposition effect

- Broad theme: when making selling decisions, we should focus on an asset's **future** performance
  - Allowing yourself to be influenced by realization utility is a mistake because it is based on the **past**
  - It leads investors to exhibit a disposition effect, which is costly, because of momentum

- Professional traders recognize this
  - basic advice is to ignore realization utility
  - part of being a disciplined trader

Selling behavior: The disposition effect

- Recent work has used techniques from neuroscience to provide evidence for realization utility
  - Frydman, Barberis, Camerer, Bossaerts, Rangel (2014)

- 28 participants traded stocks in an experimental market while we monitored their brain activity using fMRI

fMRI



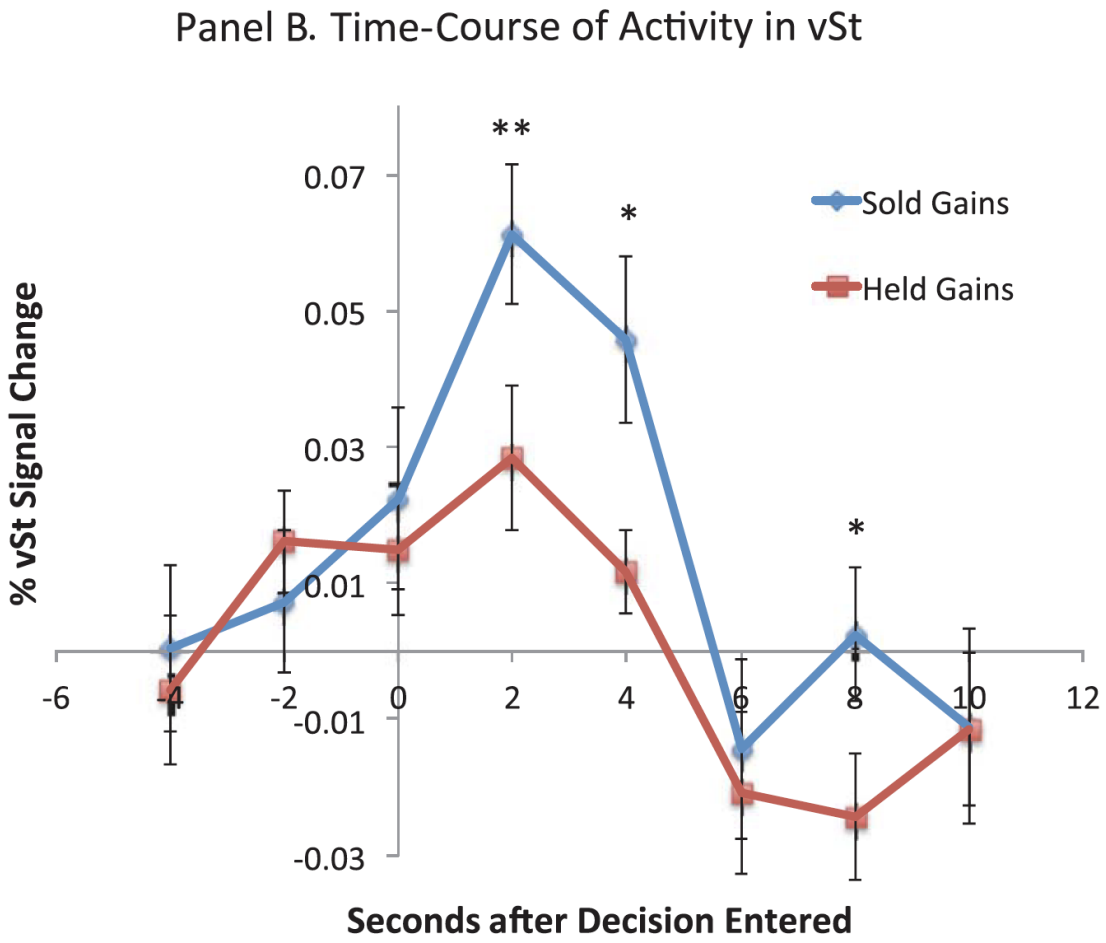
Selling behavior: The disposition effect

- The experiment was designed so that there was momentum in stocks' returns
  - and participants had enough information to figure this out
- Given this, participants should have exhibited the opposite of the disposition effect
  - but they actually exhibited a disposition effect
  - a mistake, in this context

Selling behavior: The disposition effect

- We can test the hypothesis that realization utility is driving the trading mistake
  - A part of the brain called the ventral striatum (vSt) tracks feelings of pleasure and pain
  - Under realization utility, people should feel a burst of pleasure when they sell at a gain, as compared to when they hold a stock with a similar gain.
  - When participants sell at a gain, we should see a spike in activity in the vSt, as compared to when holding a gain.
  - That is exactly what we find.

Selling behavior: The disposition effect



Selling behavior: The disposition effect

- The disposition effect can be used to extract information from the sales of corporate insiders
- One might have thought that sales of stock by insiders would be followed by poor returns
  - but the evidence for this is weak
- However, a sale of stock *at a loss* by an insider *is* followed by poor returns (Kelly, 2014)
  - intuition?

Buying behavior

- The propensity to **buy** a stock is positively related to the stock's past returns, going back at least three years
  - Such buying behavior may not be fully rational
  - Long-term prior winners subsequently underperform (De Bondt and Thaler, 1985)
- Behavioral approach: *over-extrapolation*

- Note:
  - Both buying **and** selling propensities are positively related to past returns

Buying behavior: Repurchase effect:

- Investors are more (less) reluctant to repurchase a stock if its price has gone up (down).
  - Strahilevitz, Barber, and Odean (2011)
- Interpretation: Regret
  - painful to see the price rising after selling it. To avoid being reminded over this decision, the investor chooses to mentally distance himself from the stock by not repurchasing it.
- Other applications of regret
  - Firms pay dividends (Shefrin and Statman 1984)
  - Disposition effect (Shefrin and Statman 1985)
  - Narrow framing (Barberis, Huang, and Thaler 2006)

Buying behavior: Repurchase effect:

- Experiment Frydman and Camerer (2016)
  - 92% of the subjects exhibit a repurchase effect
  - When a subject sees a price increase for a sold stock, his/her brain activity is consistent with a regret signal.
  - Those subjects who exhibit stronger neural regret signals are the same subjects who exhibit a greater repurchase effect.

- Repurchase effect and disposition effect
  - Investors with a repurchase effect also have a disposition effect
  - There might be a common psychological mechanism

Assets: Post-retirement

- Under-annuitization: According to simple models of rational behavior, people should annuitize a large fraction of their wealth at retirement
  - This insures them against **longevity risk**
- In practice, however, annuities are not very popular
- Rational approaches?
- Behavioral approach: **loss aversion**
  - People think of an annuity as an investment
  - If they pass away sooner than expected, the investment will have been a losing one

Liabilities

- Household debt has increased dramatically over time, from 50% of GDP in 1970 to 100% of GDP in 2012
  - almost all household debt is mortgage debt

- Topics that may benefit from behavioral approaches:
  - amount of debt accumulated by households (present bias)
  - mortgage choice, ARM vs. FRM (salience)
  - refinancing behavior (inattention, procrastination)
  - default decisions (inattention, inertia, feelings)
  - simultaneous presence of outstanding debt and liquid assets / retirement assets (mental accounting)
  - payday loans (instant gratification, financial literacy)

Age and financial outcomes

- There are two kinds of intelligence that are important for making good financial decisions
  - **"fluid" intelligence**: the ability to handle novel cognitive tasks
  - **"crystalized" intelligence**: a person's accumulated knowledge and experience
- Fluid intelligence is measured through specific tests
- It decreases steadily as we age

Age and financial outcomes

Figure 1. Examples of Memory and Analytical Tests

Memory

Study the following words and then write as many as you can remember

Goat
Door
Fish
Desk
Rope
Lake
Boot
Frog
Soup
Mule

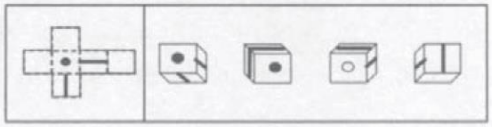
Reasoning

Select the best completion of the missing cell in the matrix



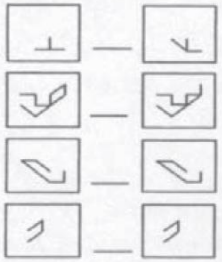
Spatial Visualization

Select the object on the right that corresponds to the pattern on the left



Perceptual Speed

Classify the pairs as same (S) or different (D) as quickly as possible



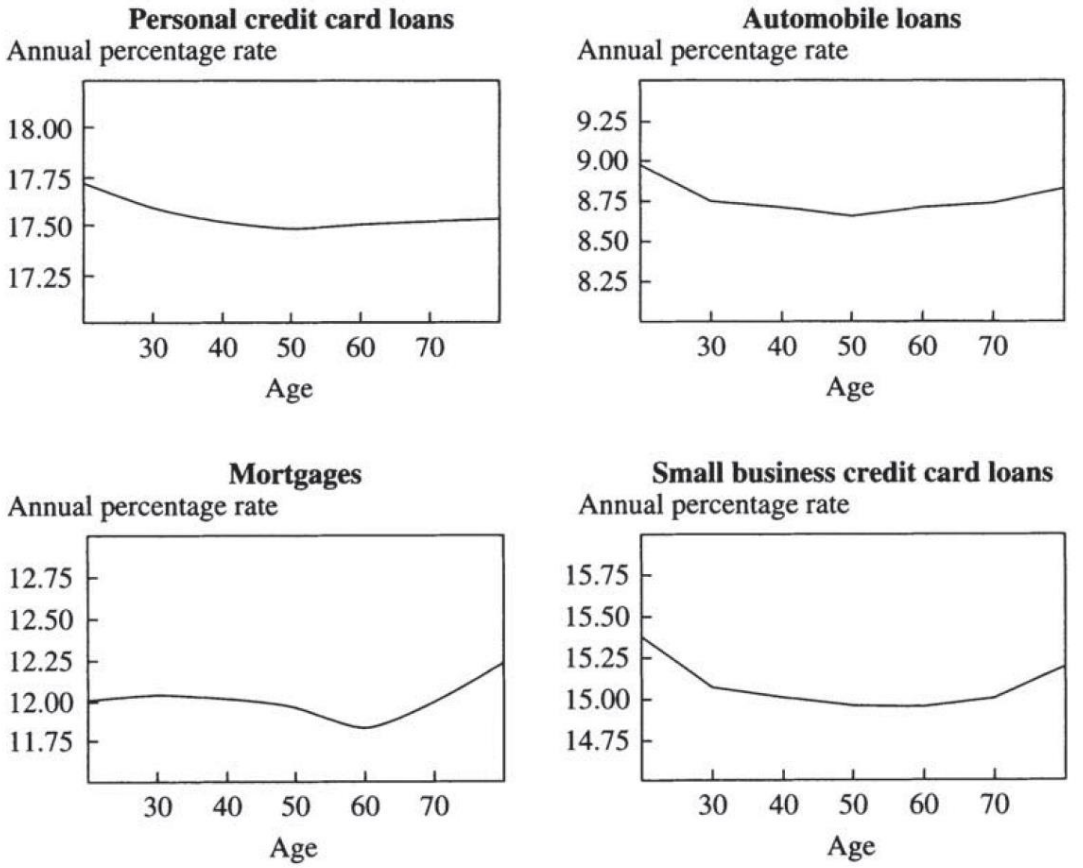
Source: Salthouse (forthcoming). Reprinted with permission.

Age and financial outcomes

- Crystalized intelligence increases as we get older, but at a slower rate over time
- Overall, this suggests that the **middle-aged** will make the best financial decisions
- The evidence is consistent with this
  - across multiple decisions, the middle-aged do better
  - e.g. in interest rates obtained on loans, or in avoiding unnecessary fees

Age and financial outcomes

Figure 9. Interest Rates Paid by Type of Loan and Age of Borrower



Summary

- The accumulated evidence points to a major problem facing society today
- Millions of households are being asked to make important financial decisions
  - yet many seem ill-equipped to do so
  - all the while, financial markets and products are becoming increasingly complex
- Crucial question: how can we help people make better financial decisions?